

Practical No. 15

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Class : CO - B
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Aim : Create Software Project scheduling charts using CRM (Critical Path Method) / PERT (Project Evaluation and Review Technique).

Objectives:

- To identify activities involved in a software project.
- To determine the dependency between project activities.
- To calculate project completion time using CPM.
- To calculate expected activity time using PERT.
- To identify the critical path and project duration.

Software / Tools Required: Any spreadsheet software or project scheduling tool. The calculations can also be performed manually.

1. Theory

CPM (Critical Path Method): CPM is a project scheduling technique used to determine the longest sequence of dependent activities. This sequence is called the critical path. Activities on the critical path have zero total float, and delay in any critical activity can delay the entire project.

PERT (Project Evaluation and Review Technique): PERT is used when activity durations are uncertain. It uses three time estimates: Optimistic time (O), Most Likely time (M), and Pessimistic time (P).

PERT Expected Time Formula: $Te = (O + 4M + P) / 6$

2. Project Activity Data

Activity	Description	Predecessor	O	M	P	Expected Time (Te)
A	Project Planning	—	2	3	4	3.00
B	Requirement Analysis	A	3	4	5	4.00
C	System Design	B	2	3	4	3.00
D	Database Design	B	2	3	4	3.00
E	Coding	C, D	5	7	9	7.00
F	Testing	E	3	4	5	4.00
G	Deployment	F	1	2	3	2.00

3. PERT Calculation

For Activity A:

$$Te = (O + 4M + P) / 6 = (2 + 4(3) + 4) / 6 = 18 / 6 = 3 \text{ days}$$

Similarly, the expected time for each activity is calculated and shown in the activity table above.

4. CPM Network Schedule

The activity dependency is: $A \rightarrow B \rightarrow C$ and $D \rightarrow E \rightarrow F \rightarrow G$. Activities C and D can be performed in parallel after completion of B. Activity E starts only after both C and D are completed.

Activity	Duration (days)	ES	EF	LS	LF	Total Float
A	3	0	3	0	3	0
B	4	3	7	3	7	0
C	3	7	10	7	10	0
D	3	7	10	7	10	0
E	7	10	17	10	17	0
F	4	17	21	17	21	0
G	2	21	23	21	23	0

ES: Earliest Start **EF:** Earliest Finish **LS:** Latest Start **LF:** Latest Finish

5. Critical Path

Critical Path = $A \rightarrow B \rightarrow C \rightarrow E \rightarrow F \rightarrow G$

The duration of the critical path is $3 + 4 + 3 + 7 + 4 + 2 = 23$ days.

6. Project Schedule Chart

Activity	Day 0–3	Day 3–7	Day 7–10	Day 10–17	Day 17–21	Day 21–23
A	■ ■ ■					
B		■ ■ ■ ■				
C			■ ■ ■			
D			■ ■ ■			
E				■ ■ ■ ■ ■ ■ ■		
F					■ ■ ■ ■	
G						■ ■

7. Result

The software project schedule was prepared using CPM and PERT techniques. The expected duration of the project is **23 days**. The critical path is **$A \rightarrow B \rightarrow C \rightarrow E \rightarrow F \rightarrow G$** . Activities on the critical path must be completed on time to avoid delay in the overall project.

8. Conclusion

CPM helps in identifying the critical activities and determining the minimum project completion time, while PERT helps in estimating activity durations when there is uncertainty. Both techniques are useful for planning, scheduling, monitoring and controlling software projects.